

# CIE-1 PYQ Bank: Model Answers

## Handwritten Question Bank & Solutions

**Syllabus Topics Covered:**

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## Unit 1 Examination Questions & Answers

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### UNIT 1

Verified model solutions for Unit 1 past exam questions.

## Unit 1 8 Marks High Priority

Ramaiah CIE-1 2025 Q1.a (8M)SEE Feb/Mar 2025(O) Q1.a (10M)SEE Jan 2026 Q1.b (6M)

Theory Link: Unit 1 § 1

Define Software Engineering. How is it more than 'just programming'? Consider a client has a vague idea of their requirements and expects them to change frequently. Which process model would you recommend and why?

Practice Problem:

Model Exam Answer

Solution:

**Software Engineering** is an engineering discipline concerned with all aspects of software production from the earliest stages of system specification through to maintaining and evolving the system after it has gone into use (Sommerville, Ch. 1).

**Why More Than Programming:**

- **Systematic Lifecycle:** Programming is merely the coding phase. Software engineering encompasses requirements analysis, architectural design, verification & validation, deployment, configuration management, and ongoing software evolution.
- **Socio-Technical & Economic Constraints:** Professional software must be developed on schedule, within organizational budgets, and adhere to strict quality attributes (maintainability, dependability, efficiency, and usability).
- **Team-Based & Industrial Scale:** Programming is often individual or small-scale; software engineering provides the organizational framework, version control, automated testing, and governance needed for large teams building complex, long-lived systems.

**Recommendation Case:**

**Scenario:** Client with vague, emerging requirements expecting frequent changes.

**Recommended Model:** Agile Incremental Development / Extreme Programming (XP) or Scrum (or Evolutionary Prototyping).

**Justification:**

## Unit 1 8 Marks High Priority

Ramaiah CIE-1 2025 Q2.a (8M) SEE Feb/Mar 2025 Q1.a (8M) SEE Feb 2024 Q1.c (8M) SEE Jan 2026 Q1.a (8M)

Theory Link: Unit 1 § 4

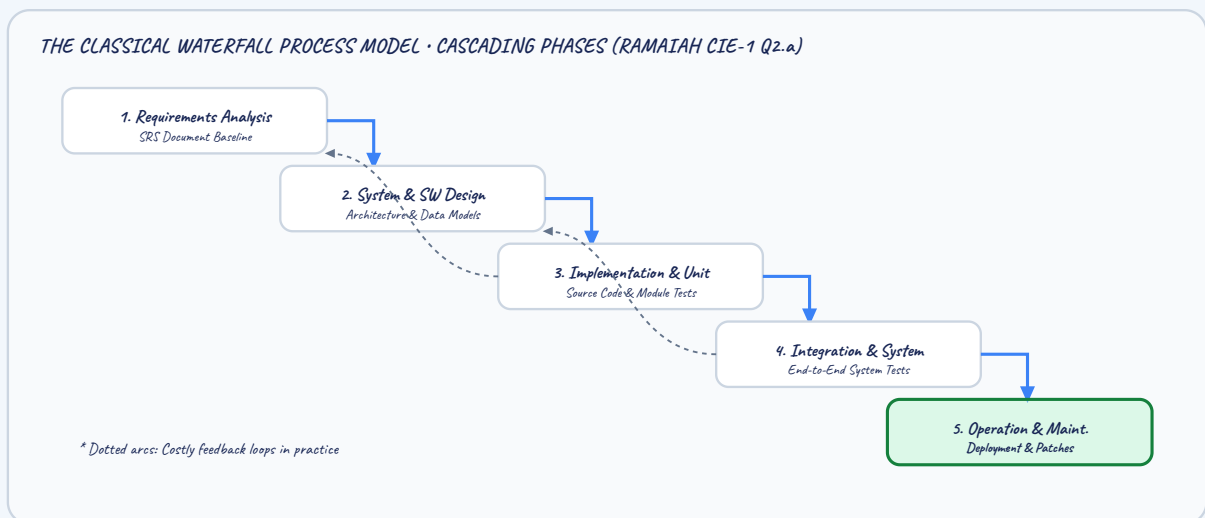
With a neat diagram, explain in detail about the Waterfall process model for software development. Identify its advantages and limitations.

Practice Problem:

Model Exam Answer

Solution:

The **Waterfall Model** (Classic Life Cycle) is a plan-driven software process model where each fundamental process activity is organized into separate, sequential phases, with the output of one phase serving as the formal baselined input to the next.



**Phases:**

- **1. Requirements Definition:** System services, constraints, and goals are established in consultation with stakeholders and formally documented in an SRS.
- **2. System and Software Design:** Allocates requirements to hardware/software systems; establishes overall software architecture and data structures.
- **3. Implementation and Unit Testing:** Software design is realized as sets of programs or program units; verified through unit testing.
- **4. Integration and System Testing:** Individual program units are

## Unit 1 6 Marks High Priority

Ramaiah CIE-1 2025 Q3.a (6M) SEE Feb/Mar 2025 Q2.a (6M) SEE Jan 2026 Q2.a (8M) SEE Feb 2024 Q2.b (6M) SEE Feb/Mar 2025(O) Q1.b (10M)

Theory Link: Unit 1 § 4

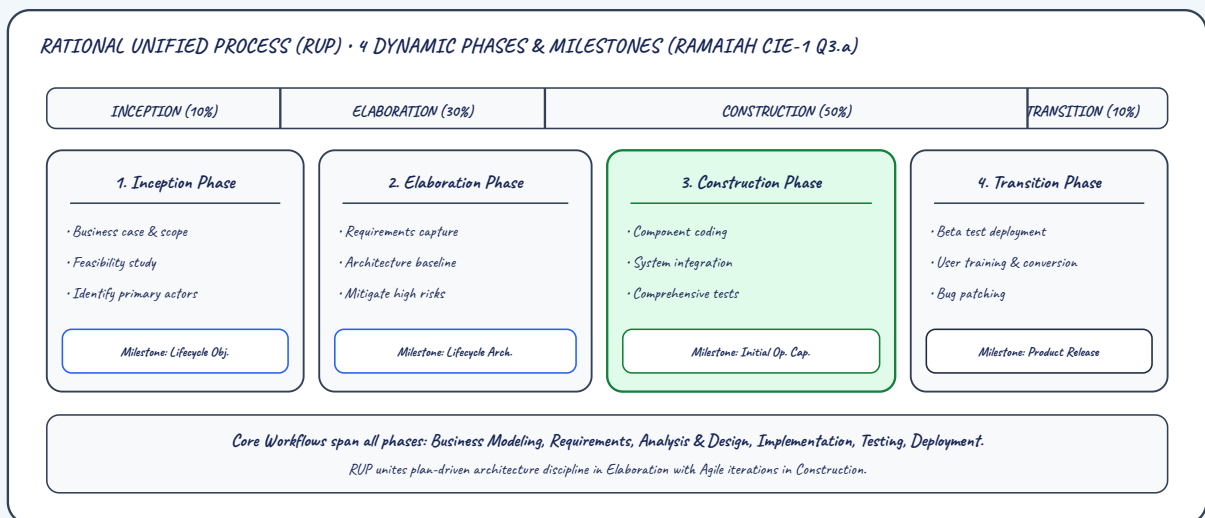
Describe and elaborate the four distinct phases of the Rational Unified Process (RUP), with a neat diagram.

Practice Problem:

## Model Exam Answer

Solution:

The **Rational Unified Process (RUP)** is a modern generic process framework that combines elements of all generic process models. It has two dimensions: dynamic (lifecycle phases over time) and static (workflows/disciplines across iterations).

**Four Phases:**

- **1. Inception Phase:** Establish the business case, project scope, identify all external entities (actors), define major use cases at a high level, and assess business viability and financial risks.
- **2. Elaboration Phase:** Deepen understanding of problem domain; establish an architectural baseline; identify and eliminate the highest project risks; produce the executable architecture.
- **3. Construction Phase:** System design, programming, and unit/integration testing. Components are implemented, integrated into the architecture, and tested against use cases; produces the first external beta release.

Unit 1 7 Marks High Priority

SEE Feb/Mar 2025 Q1.b (7M) SEE Jan 2026 Q2.b (6M) SEE Feb 2024 Q2.a (8M) SEE Feb/Mar 2025(0) Q2.a (10M)

Theory Link: Unit 1 § 7

Compare and contrast plan-driven processes and agile processes with a suitable use case. Demonstrate how they can be combined in a hybrid model.

Practice Problem:

Model Exam Answer

Solution:

Comparison Table:

Headers:

- Attribute
- Plan-Driven Development
- Agile Development

Rows:

- ['Philosophy', 'Anticipate changes upfront; extensive formal planning and architecture.', 'Embrace change; adapt iteratively through customer collaboration.']
- ['Documentation', 'Comprehensive, heavy documentation (SRS, SDD, traceability matrices).', 'Working software prioritized over comprehensive documentation; user stories.']
- ['Customer Involvement', 'Formal involvement during requirements and acceptance sign-offs.', 'Continuous, embedded customer collaboration throughout development.']
- ['Testing Strategy', 'Formal V-Model testing following implementation phase.', 'Test-Driven Development (TDD) and continuous automated integration.']
- ['Team Structure', 'Hierarchical, specialized functional teams with strict governance.', 'Cross-functional, self-organizing teams with high tacit communication.']

## Unit 1 6 Marks High Priority

SEE Feb/Mar 2025 Q2.c (6M) SEE Jan 2026 Q2.c (6M) SEE Feb 2024 Q2.c (6M)

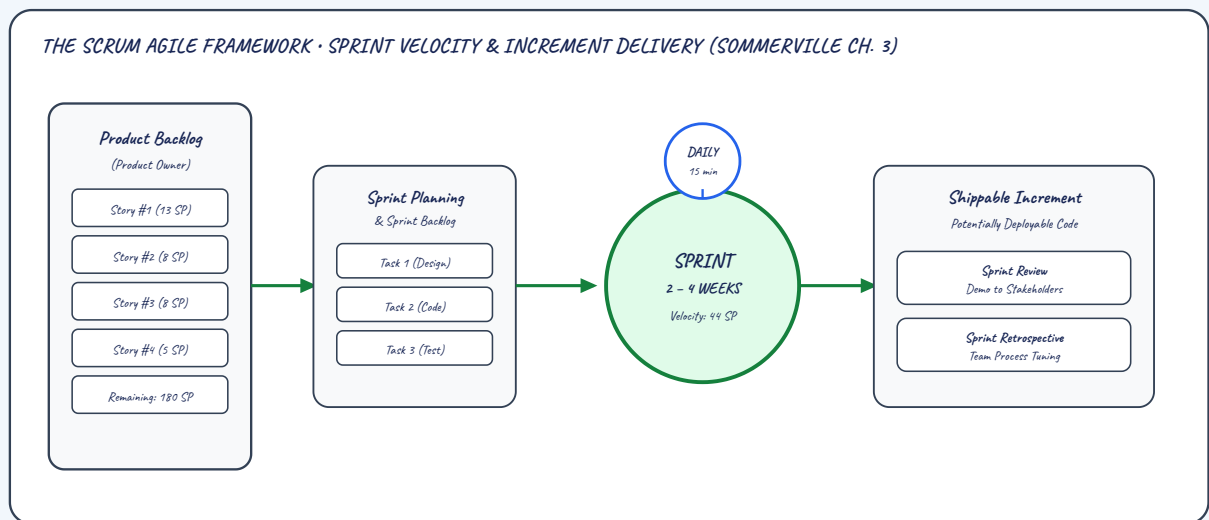
Theory Link: Unit 1 § 7

What is the need for Extreme Programming (XP)? Draw and describe the Extreme Programming release/iteration cycle, highlighting key XP practices.

Practice Problem:

Model Exam Answer

Solution:



**Need For Xp:**

Traditional methods assumed changes late in development were prohibitively expensive. **Extreme Programming (XP)** was created by Kent Beck to take recognized good development practices (e.g. testing, reviews, integration) and push them to 'extreme' levels to make software development highly responsive to customer needs while maintaining code quality.

**Xp Cycle Description:**

- **1. User Stories:** Customer requirements are written on story cards by the customer.
- **2. Release Planning:** Team and customer prioritize stories into the next release schedule and break stories into developmental tasks.
- **3. Iteration Development:** In 1-2 week cycles, developers implement tasks using Pair Programming and Test-Driven Development.
- **4. Acceptance Tests:** Customer-defined automated tests verify completed



## Unit 1 6 Marks Medium Priority

SEE Feb/Mar 2025 Q1.c (5M) SEE Jan 2026 Q1.c (6M) SEE Jan 2023 Q1 (5M)

[Theory Link: Unit 1 § 2](#)

List out the different ethical responsibilities of the software engineering profession.  
State the eight principles of the ACM/IEEE-CS Code of Ethics.

Practice Problem:

Model Exam Answer

Solution:

Software engineers must behave in an honest and ethically responsible manner, not merely follow the law. Technical competence must be joined with ethical integrity because software systems directly impact public safety, privacy, and economic well-being.

Professional Responsibilities:

- **Confidentiality:** Respect client/employer confidentiality regardless of whether a formal non-disclosure agreement was signed.
- **Competence:** Do not misrepresent level of competence; do not knowingly accept work outside personal competence.
- **Intellectual Property Rights:** Respect patents, copyrights, and open-source licenses; never use unauthorized software.
- **Computer Misuse:** Never use technical abilities to breach security, deploy malware, or access private systems unauthorized.

Eight Principles:

- **1. PUBLIC:** Act consistently with the public interest as paramount.
- **2. CLIENT AND EMPLOYER:** Act in a manner that is in the best interests of client and employer, consistent with public interest.
- **3. PRODUCT:** Ensure products and related modifications meet the highest professional standards possible.
- **4. JUDGMENT:** Maintain integrity and independence in professional judgment.
- **5. MANAGEMENT:** Managers and leaders shall subscribe to and promote an ethical approach to software management.
- **6. PROFESSION:** Advance the integrity and reputation of the profession consistent with public interest.
- **7. COLLEAGUES:** Be fair to and supportive of colleagues.



## Unit 1 8 Marks Medium Priority

SEE Jan 2026 Q1.a (8M) SEE Jan 2023 Q1 (8M)

Theory Link: Unit 1 § 5

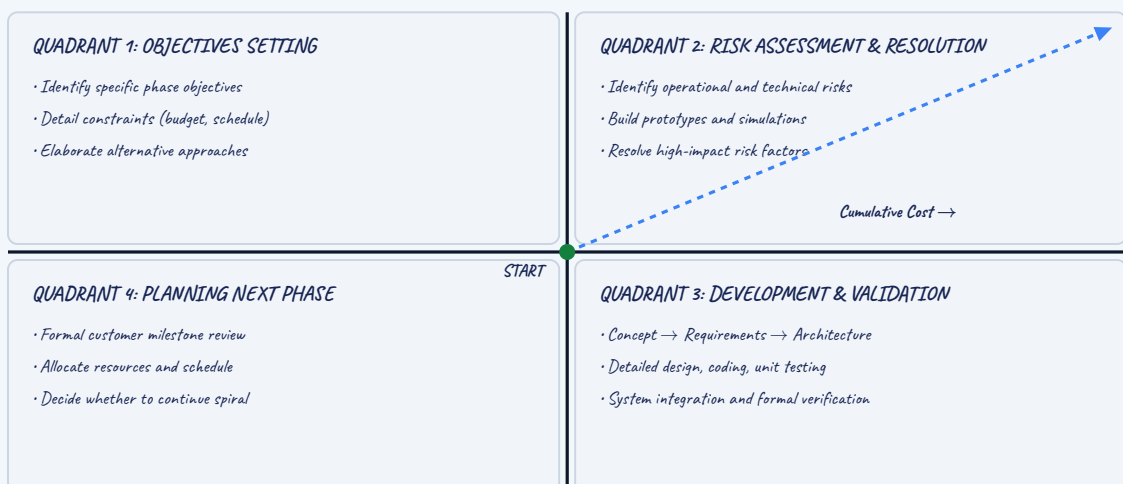
With a neat diagram explain Boehm's Spiral Model. Discuss its risk-driven nature, four quadrants, and its advantages over the waterfall model.

Practice Problem:

## Model Exam Answer

Solution:

**Boehm's Spiral Model** is a risk-driven software process framework where the development lifecycle is represented as a spiral, rather than a sequence of activities with backtracking. Each loop represents a phase of the process.

**Four Quadrants:**

- **1. Objective Setting (Quadrant 1):** Specific objectives for the phase are identified. Constraints on processes and products are defined, and a detailed management plan is drawn up.
- **2. Risk Assessment and Reduction (Quadrant 2):** For each identified project risk, a detailed analysis is carried out. Steps are taken to reduce risks (e.g. prototyping, simulations, benchmarking).
- **3. Development and Validation (Quadrant 3):** A development model for the system is chosen based on remaining risks (e.g. waterfall if low risk, evolutionary prototype if UX risk, etc.). The increment is built and validated.

## Unit 1 6 Marks Medium Priority

SEE Feb 2024 Q1.a (6M)

Theory Link: Unit 1 § 3

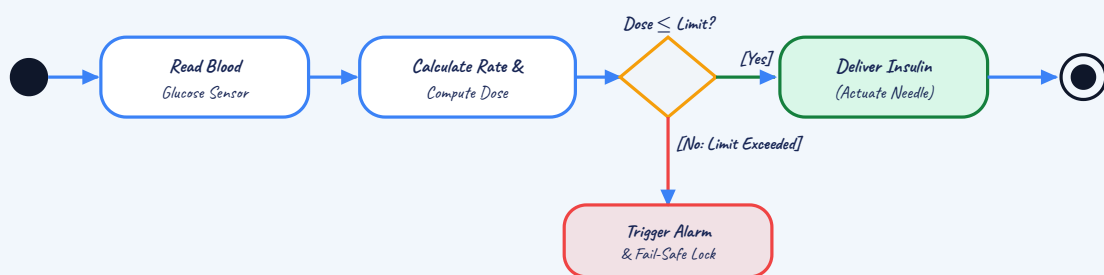
With a neat diagram, explain the activity model of the Insulin Pump control system.  
Why is this considered a safety-critical system?

Practice Problem:

Model Exam Answer

Solution:

The **Insulin Pump** is a safety-critical embedded system used to regulate blood sugar in patients with Type 1 diabetes. It measures micro-current blood glucose levels, computes insulin requirement, and drives a pump micro-motor.

Activity Steps:

- 1. Read blood sensor output.
- 2. Calculate blood sugar level and rate of change.
- 3. Compute required insulin dose.
- 4. Run safety check: ensure dose  $\leq$  maximum safe single bolus.
- 5. Command needle pump actuator to deliver computed units.
- 6. Log delivery timestamp and insulin remaining.
- 7. Wait for scheduled sampling interval (e.g. 10 minutes).

It is safety-critical because failure can cause death: delivering too little insulin leads to hyper-glycaemic coma; delivering an overdose leads to fatal hypoglycaemia. The system must never fail silently and must run fail-safe self-diagnostics.

**Source:** Sommerville 10e, Chapter 1.4 (pp. 31-34)

## Unit 2 Examination Questions & Answers

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### UNIT 2

Verified model solutions for Unit 2 past exam questions.

## Unit 2 7 Marks High Priority

Ramaiah CIE-1 2025 Q1.b (7M) SEE Feb/Mar 2025 Q3.b (6M) SEE Feb/Mar 2025 Q4.b (6M) SEE Jan 2026 Q3.a (6M) SEE Feb 2024 Q3.a (6M)

Theory Link: Unit 2 § 1

Differentiate between Functional and Non-Functional Requirements (NFRs).

Provide two examples of each for an e-commerce website and develop a list of NFRs for a real-time banking application.

Practice Problem:

Model Exam Answer

Solution:

**Definitions:**

**Functional:** **Functional Requirements** describe the specific services, functions, and behaviors the system must provide, detailing how the system should react to particular inputs and how it should behave in specific situations.

**Non Functional:** **Non-Functional Requirements (NFRs)** are constraints on the services or functions offered by the system. They include performance constraints, reliability, availability, security, regulatory standards, and development toolchain constraints.

**Comparison Table:**

**Headers:**

- Dimension
- Functional Requirements
- Non-Functional Requirements

**Rows:**

- ['Scope', 'Individual features or sub-functions.', 'System-wide properties and emergent characteristics.']
- ['Focus', 'What the system should DO.', 'How well or under what constraints the system PERFORMS.']
- ['Verification', 'Tested via functional/unit test cases.', 'Tested via load, stress, security, and benchmark tests.']

## Unit 2 7 Marks High Priority

Ramaiah CIE-1 2025 Q2.b (7M) SEE Feb 2024 Q4.b (6M) SEE Jan 2026 Q3.c (6M) SEE Feb/Mar 2025(O) Q4.b (8M)

Theory Link: Unit 2 § 3

What are the common challenges faced during requirements elicitation, and how can a software engineer mitigate them?

Practice Problem:

Model Exam Answer

Solution:

**Requirements Elicitation** is the process of interacting with stakeholders to discover what services the system should provide, the required performance, and system constraints.

**Challenges And Mitigations:**

- {'challenge': '**1. Articulation Gap (Stakeholders don't know what they want):** Users find it difficult to express what they need, except in the most general terms, or express unrealistic demands.', 'mitigation': 'Develop rapid interactive throwaway prototypes, use concrete scenarios and User Stories rather than abstract questionnaires.'}
- {'challenge': '**2. Domain Jargon & Communication Barrier:** Stakeholders describe needs using domain-specific jargon that software engineers misunderstand, and engineers use technical jargon that confuses users.', 'mitigation': 'Establish a shared project glossary/data dictionary; conduct joint workshop sessions (JAD) with embedded domain experts.'}
- {'challenge': '**3. Conflicting Stakeholder Requirements:** Different departments (e.g. Finance vs. Security vs. Operations) have incompatible goals and priorities.', 'mitigation': 'Conduct requirements negotiation workshops; use MoSCoW prioritization (Must have, Should have, Could have, Won't have) and formal trade-off matrices.'}
- {'challenge': '**4. Tacit Knowledge & Social Factors:** Users perform routine complex organizational procedures unconsciously that they forget to mention; organizational politics can skew input.', 'mitigation': 'Employ **Ethnography** (observing users at work in their real environment) to uncover tacit workflow patterns and institutional culture.'}
- {'challenge': '**5. Requirement Volatility:** Business priorities, regulations, and

## Unit 2 5 Marks High Priority

Ramaiah CIE-1 2025 Q3.b (5M) SEE Jan 2026 Q5.a (8M) SEE Feb 2024 Q5.b (10M)

Theory Link: Unit 2 § 4

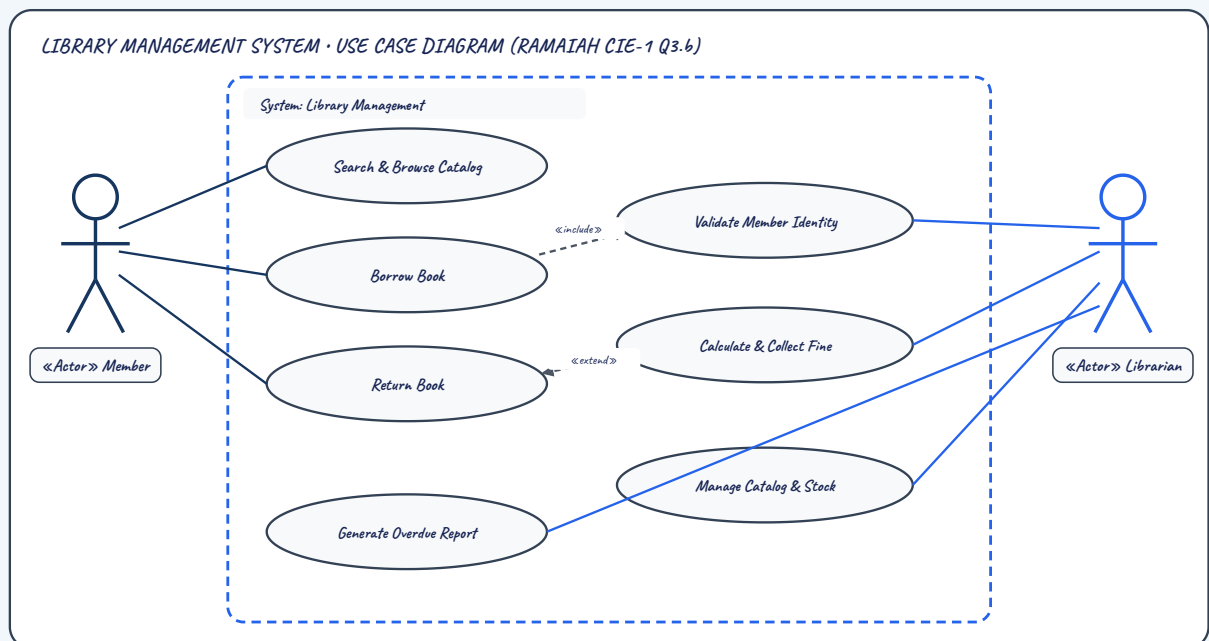
Create a simple Use Case Diagram for a 'Library Management System' involving actors like Member and Librarian, with at least three key use cases, and explain include/extend relationships.

Practice Problem:

## Model Exam Answer

Solution:

A **Use Case Diagram** in UML models the functional requirements of a system by identifying interactions between external actors and system use cases.

**Actors:**

- **Member:** Student/Faculty borrowing books, searching catalog, reserving items.
- **Librarian:** Staff managing book catalog, issuing/returning books, assessing fines.

**Use Cases:**

1. **Search Catalog:** Both Member and Librarian query library database by title/author/ISBN.
2. **Borrow / Issue Book:** Librarian issues book to verified Member.



## Unit 2 7 Marks High Priority

Ramaiah CIE-1 2025 Q3.c (4M)SEE Feb/Mar 2025 Q3.c (7M)SEE Feb 2024 Q3.c (6M)SEE Feb/Mar 2025(O) Q3.c (6M)SEE Jan 2023 Q4.b (5M)

Theory Link: Unit 2 § 3

List and briefly explain the requirement validation process and the five different checks carried out during validation.

Practice Problem:

Model Exam Answer

Solution:

**Requirements Validation** is the process of checking that requirements actually define the system that the customer really wants. Discovering requirements errors at this stage saves astronomical rework costs later in the development cycle.

**Five Checks:**

- **1. Validity Checks:** Does the system provide the functions which best support the customer's real operational needs?
- **2. Consistency Checks:** Requirements in the document should not conflict (e.g. one section stating 24/7 availability while another stating nightly 4-hour maintenance shutdown).
- **3. Completeness Checks:** The document should include all requirements and constraints requested by the user, covering all normal and exception cases.
- **4. Realism Checks:** Can the requirements actually be implemented using known technologies, within budget and schedule constraints?
- **5. Verifiability Checks:** Can the requirements be objectively tested? Every requirement must be phrased so that a software tester can design an unambiguous pass/fail test.

**Validation Techniques:**

- **Requirements Reviews (Walkthroughs):** Systematic manual analysis by developers, customers, and contractors.
- **Prototyping:** Building an executable model to test user reaction and validate assumptions.
- **Test-Case Generation:** If creating a test case is impossible for a requirement. it is ambiguous and must be rewritten.



## Unit 2 7 Marks High Priority

SEE Feb/Mar 2025 Q3.a (7M) SEE Feb 2024 Q4.c (8M)

Theory Link: Unit 2 § 3

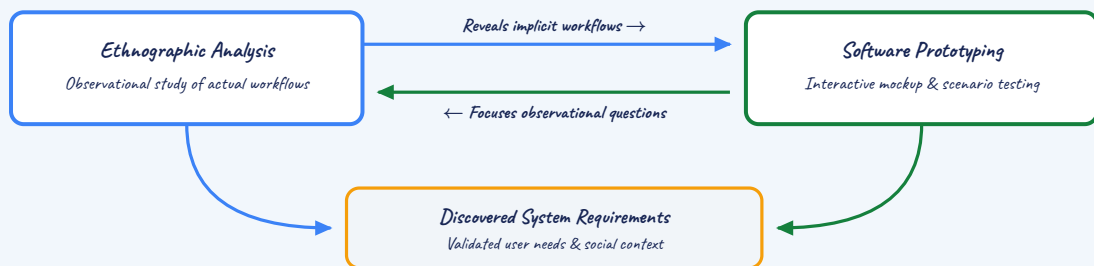
What is Ethnography? Illustrate with a diagram how combining Ethnography and Prototyping helps the requirements discovery process.

Practice Problem:

## Model Exam Answer

Solution:

**Ethnography** is an observational technique where an analyst immerses themselves in the actual working environment where the system is to be used, watching the day-to-day work and noting how people actually collaborate and use information.

Why Ethnography Matters:

- Discovers **implicit (tacit) requirements** that workers cannot articulate because procedures are second nature to them.
- Identifies **social and organizational factors** that determine whether a software system will actually be accepted or resisted in practice.
- Limitation: Ethnography reveals how people currently work, but cannot innovate or establish new organizational structures.

Combining With Prototyping:

Prototyping is focused and forward-looking, but can overlook subtle workplace dynamics. When combined:

- **Ethnography** informs the early prototype by grounding it in real workplace habits.
- **Prototyping** directs the ethnographer's focus to specific workflows and provides a concrete artifact for users to critique.

Unit 2 7 Marks Medium Priority

SEE Feb/Mar 2025 Q4.a (7M) SEE Feb/Mar 2025(O) Q3.b (6M)

Theory Link: Unit 2 § 2

What are the problems with using Natural Language for specifying systems requirements? Explain how Structured Natural Language specifications overcome these problems.

Practice Problem:

Model Exam Answer

Solution:

*Problems With Natural Language:*

- **1. Ambiguity & Lack of Clarity:** Authors mistakenly assume readers share their implicit context, causing words to be interpreted differently by developers and clients.
- **2. Requirements Confusion:** Functional requirements, non-functional requirements, project goals, and design details become mixed up in the same paragraph.
- **3. Requirements Amalgamation:** Multiple distinct requirements are lumped together into a single sprawling sentence.
- **4. Over-Flexibility:** The same concept can be phrased in ten different ways, making automated verification or cross-referencing impossible.

*Structured Natural Language Solution:*

- Uses a standard form/template with mandatory predefined fields to write requirements.
- **Standard Fields:** Function Name, Description, Inputs (source & format), Outputs (destination & rate), Pre-conditions, Post-conditions, Side-effects, Exceptions.
- Maintains readability for non-technical clients while enforcing rigor, completeness, and consistency for engineers.

**Source:** Sommerville 10e, Chapter 4.3 (pp. 112-116)

Unit 2 7 Marks High Priority

SEE Feb/Mar 2025 Q4.c (7M) SEE Jan 2026 Q3.b (8M) SEE Jan 2023 Q4.a (10M) SEE Feb/Mar 2025(O) Q3.b (6M)

Theory Link: Unit 2 § 2

Explain the structure of a Software Requirements Specification (SRS) document according to IEEE 830 standard and highlight the characteristics of a good SRS document.

Practice Problem:

Model Exam Answer

Solution:

The **Software Requirements Specification (SRS)** is the official statement of what the system developers should implement. It serves as a contract between customer and developer.

*Ieee 830 Structure:*

- **1. Introduction:** 1.1 Purpose, 1.2 Scope, 1.3 Definitions/Acronyms, 1.4 References, 1.5 Overview.
- **2. Overall Description:** 2.1 Product Perspective, 2.2 Product Functions, 2.3 User Characteristics, 2.4 General Constraints, 2.5 Assumptions and Dependencies.
- **3. Specific Requirements:** Detailed functional requirements, non-functional requirements (performance, reliability, security), external interface requirements (user, hardware, software, communications), and design constraints.
- **4. Appendices & Index:** Hardware specs, sample input/output forms, glossary.

*Characteristics Of Good Srs:*

- **Correct:** Every requirement stated is one that the software shall meet.
- **Unambiguous:** Every requirement has only one interpretation.
- **Complete:** Includes all significant requirements (inputs, outputs, exceptions).
- **Consistent:** No two requirements conflict with each other.
- **Verifiable:** There exists a finite, cost-effective test to verify each requirement.
- **Modifiable & Traceable:** Well-structured with unique IDs so changes can

## Unit 2 10 Marks High Priority

SEE Jan 2026 Q4.a (10M) SEE Feb 2024 Q1.b (6M) SEE Feb 2024 Q4.a (6M)

Theory Link: Unit 2 § 3

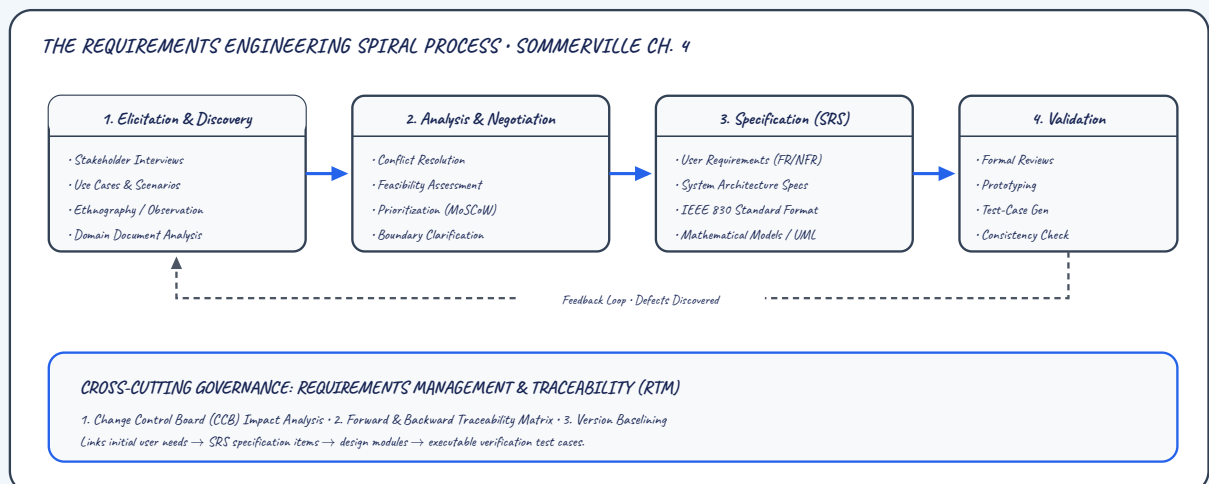
With a neat diagram, explain the overall Requirements Engineering (RE) process and its primary activities.

Practice Problem:

## Model Exam Answer

Solution:

The **Requirements Engineering Process** is a structured set of activities used to discover, analyze, specify, validate, and maintain a software requirements document.

**Four Primary Activities:**

- **1. Feasibility Study:** Assesses whether the system contributes to business objectives, can be engineered within current tech/budget, and integrates with existing systems. Outputs a Feasibility Report.
- **2. Requirements Elicitation and Analysis:** Working with customers to discover domain info, system services, and operational constraints. Involves discovery, classification, prioritization, and negotiation.
- **3. Requirements Specification:** Translating information into a formal user and system requirements document (SRS).
- **4. Requirements Validation:** Checking that requirements are complete, consistent, realistic, verifiable, and conform to user desires.

**Source:** Sommerville 10e, Chapter 4.4 (pp. 116-125)

## Unit 2 10 Marks Medium Priority

SEE Jan 2026 Q4.b (10M)

Theory Link: Unit 2 § 1

Consider an Automated Traffic Signal System. Write functional and non-functional requirements for this system.

Practice Problem:

Model Exam Answer

Solution:**System Context:**

An automated traffic control system installed at a busy four-way junction with inductive loop vehicle sensors, pedestrian push-buttons, and emergency vehicle optical detectors.

**Functional Requirements:**

- **FR-01:** The system shall cycle through Red, Yellow, and Green lights for each lane based on a dynamic time scheduler.
- **FR-02:** When inductive loop sensors detect a queue of > 10 vehicles in lane A, the system shall extend lane A green light duration by 15 seconds (up to a max of 60 seconds).
- **FR-03:** Upon pedestrian button press, the system shall transition conflicting vehicular signals to yellow and then red within 20 seconds, and activate the pedestrian walk indicator for 15 seconds.
- **FR-04:** Upon receiving an emergency vehicle optical strobe signal (ambulance/fire engine), the system shall immediately grant green right-of-way to that corridor while holding all conflicting lanes at red.
- **FR-05:** The system shall log vehicle counts and transition timings to local flash storage and upload hourly to central municipal servers.

**Non Functional Requirements:**

- **NFR-01 (Reliability):** MTBF shall exceed 50,000 hours; hardware failures shall automatically trip signals into a flashing yellow/red fail-safe mode.
- **NFR-02 (Safety):** The controller must never illuminate green signals for conflicting perpendicular lanes simultaneously; verified via independent hardware interlock.
- **NFR-03 (Performance):** Transition latency between emergency strobe detection and green corridor activation shall not exceed 1.0 second.

Unit 2 6 Marks Medium Priority

SEE Feb/Mar 2025(0) Q4.c (6M)

Theory Link: Unit 2 § 3

With a diagram explain the principal stages of a Change Management process in Requirements Management.

Practice Problem:

Model Exam Answer

Solution:

**Requirements Management** is the process of managing changing requirements during requirements engineering and system development. Because change is inevitable, an explicit change management process ensures controlled, cost-aware evolution.



**Three Stages:**

- **1. Problem Analysis & Change Specification:** Identifies problem or change proposal; checks that request is valid and not a duplicate.
- **2. Change Analysis & Costing:** Assesses impact on existing system, calculates cost in dollars and schedule delay using Traceability Matrices (RTM).
- **3. Change Implementation:** Modifies requirements document, design models, and source code; regression tests; updates version baselines.

**Source:** Sommerville 10e, Chapter 4.6 (pp. 126-129)



## Unit 2 6 Marks Medium Priority

SEE Jan 2026 Q5.a (8M)

[Theory Link: Unit 2 § 5](#)

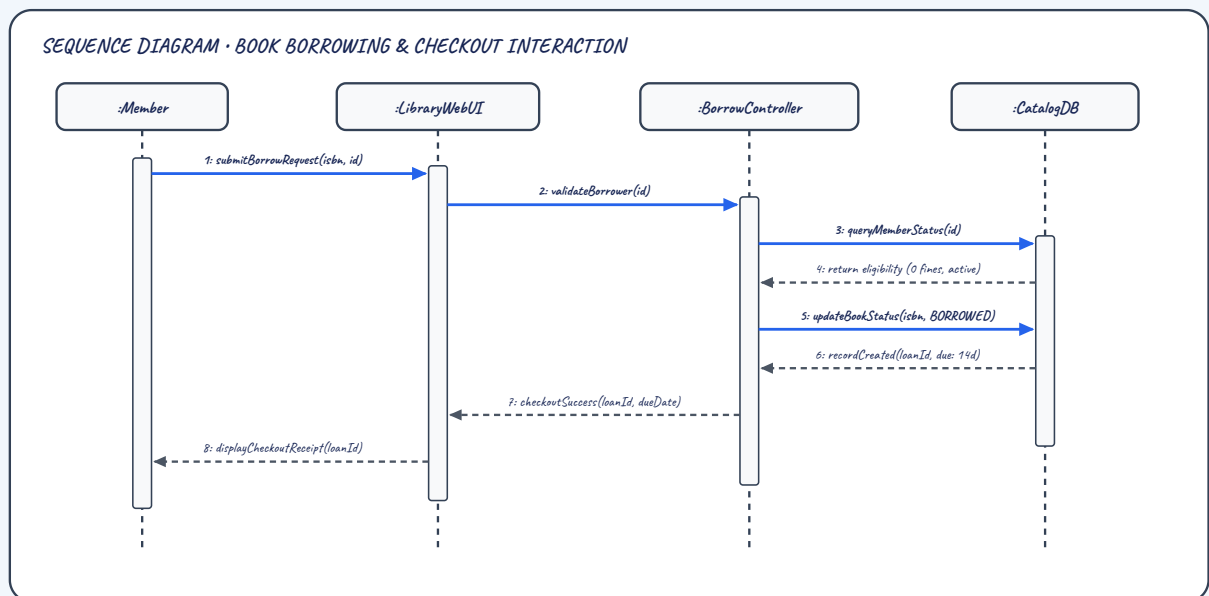
What is an Interaction Model? Explain how Sequence Diagrams model dynamic interactions with lifelines, messages, and combined fragments.

Practice Problem:

Model Exam Answer

Solution:

**Interaction Models** show how system components interact with each other and with external actors. They model dynamic system behavior.

**Sequence Diagram Elements:**

- **Lifelines:** Dashed vertical lines representing participating objects or actors across time (time progresses downwards).
- **Activation Bars:** Tall narrow rectangles placed on lifelines showing the period during which an object is executing an operation.
- **Synchronous Message (Solid Arrow with Filled Head):** Caller blocks and waits for operation to return before proceeding.
- **Asynchronous Message (Solid Line with Open Arrowhead):** Caller sends message and continues execution immediately.
- **Return Message (Dashed Line with Open Arrowhead):** Value or control returned from called operation.



## Unit 2 6 Marks Medium Priority

Sommerville 10e Unit 2 Standard Curriculum Question SEE 2023 Q5.b (12M)

Theory Link: Unit 2 § 7Differentiate between Data-Driven (Activity) models and Event-Driven (State Machine) models. Provide a simple State Machine diagram for a Microwave Oven.Practice Problem:

Model Exam Answer

Solution:

- **Data-Driven Models (Activity Diagrams):** Show the sequence of actions and processing steps involved in handling data. Best for batch processing, business data pipelines, and algorithms.
- **Event-Driven Models (State Machine Diagrams):** Show how a system responds to internal and external events by transitioning between discrete system states. Essential for real-time and embedded systems.

State Machine Elements:

- **States (Rounded Rectangles):** An abstraction of the system's condition (e.g. Waiting, Cooking, Door Open).
- **Transitions (Directed Arrows):** Movement between states triggered by an event: Trigger [Guard] / Action.

**Source:** Sommerville 10e, Chapter 5.4 (pp. 154-159)**Out of CIE-1 Scope Questions (18 Questions)****ARCHIVED**

The following questions were identified in past papers but fall outside the official CIE-1 syllabus. They are preserved here to ensure zero questions are dropped.

| Paper / Origin                        | Question Summary                                                                                                                              | Exclusion Rationale                                                                                  |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Ramaiah CIE-2<br>2025<br>(23/12/2025) | Illustrate advantages and disadvantages of the repository and layered architectural design models.                                            | <u>Architectural Patterns (Layered, Repository) are in Unit 3, which is outside the CIE-1 scope.</u> |
| Ramaiah CIE-2<br>2025<br>(23/12/2025) | Demonstrate equivalence class test cases for the NextDate function.                                                                           | <u>Software Testing / Equivalence Partitioning belongs to Unit 4, outside CIE-1 scope.</u>           |
| Ramaiah CIE-2<br>2025<br>(23/12/2025) | Explain the factors effecting software pricing.                                                                                               | <u>Project Management and Software Pricing belong to Unit 5, outside CIE-1 scope.</u>                |
| Ramaiah CIE-2<br>2025<br>(23/12/2025) | With a neat diagram, explain the architecture of a language processing system.                                                                | <u>Application Architectures belongs to Unit 3, outside CIE-1 scope.</u>                             |
| Ramaiah CIE-2<br>2025<br>(23/12/2025) | Using boundary value testing, write the test cases of an improved version of triangle problem that takes 3 integers as sides of the triangle. | <u>Boundary Value Testing belongs to Unit 4, outside CIE-1 scope.</u>                                |
| Ramaiah CIE-2<br>2025<br>(23/12/2025) | Explain why the process of project planning is iterative and why a plan must be continually reviewed during a software project.               | <u>Project Planning belongs to Unit 5, outside CIE-1 scope.</u>                                      |
| Ramaiah CIE-2<br>2025<br>(23/12/2025) | Briefly Explain any two software design patterns with suitable diagrams.                                                                      | <u>Design Patterns belong to Unit 3, outside CIE-1 scope.</u>                                        |

| Paper / Origin                        | Question Summary                                                                                                                      | Exclusion Rationale                                                                   |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Ramaiah CIE-2<br>2025<br>(23/12/2025) | Define Equivalence class testing.<br>Write weak robust equivalence class test cases for Commission Problem.                           | <u>Testing Commission Problem belongs to Unit 4, outside CIE-1 scope.</u>             |
| Ramaiah CIE-2<br>2025<br>(23/12/2025) | Discuss strategies used for risk management and managing people at a workplace in an organization.                                    | <u>Risk Management &amp; People Management belong to Unit 5, outside CIE-1 scope.</u> |
| SEE Feb/Mar<br>2025                   | Architectural styles, types of architectural styles, documenting architecture.                                                        | <u>Architectural Design is in Unit 3, outside CIE-1 scope.</u>                        |
| SEE Feb/Mar<br>2025                   | Car Rental System Object-Oriented Design principles and Class Diagram.                                                                | <u>OOD and Class-based detailed system implementation is in Unit 3.</u>               |
| SEE Feb/Mar<br>2025                   | Equivalence class test cases for triangle problem, Specification-based vs code-based, Robust worst-case BVT formula.                  | <u>Software Testing belongs to Unit 4, outside CIE-1 scope.</u>                       |
| SEE Feb/Mar<br>2025                   | Project management goals, plan-driven vs agile planning, categories of risk, human needs hierarchy, software pricing factors.         | <u>Project Management belongs to Unit 5, outside CIE-1 scope.</u>                     |
| SEE Jan 2026                          | Google Docs architectural pattern, Repository vs Client-Server, ATM software architecture, Observer Pattern, Open source development. | <u>Architectural patterns, design patterns and open source belong to Unit 3.</u>      |

| Paper / Origin     | Question Summary                                                                                                             | Exclusion Rationale                                               |
|--------------------|------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| SEE<br>Jan<br>2026 | NextDate BVT, Fault taxonomy, Triangle problem classes, Levels of testing, Black-box vs White-box.                           | <u>Software Testing belongs to Unit 4, outside CIE-1 scope.</u>   |
| SEE<br>Jan<br>2026 | Project risk management, plan-driven development, software metrics, people conflict management, software pricing strategies. | <u>Project Management belongs to Unit 5, outside CIE-1 scope.</u> |
| SEE<br>Feb<br>2024 | Architectural styles, component design principles, ATM system architecture.                                                  | <u>Architectural design belongs to Unit 3.</u>                    |
| SEE<br>Feb<br>2024 | Verification vs Validation, automated unit testing, re-engineering, hardware vs software test engineer, lifecycle defects.   | <u>Software testing &amp; evolution belongs to Unit 4.</u>        |